# Device User Behaviour Analysis Report

## Project Overview

This project analyzes mobile device usage patterns across 700 users to understand how device type, operating system, and user demographics influence application usage, screen-on time, and overall device behavior.

The goal of the analysis is to identify usage trends, behavioral segments, and engagement patterns that could help product teams, mobile app developers, and marketers better understand how users interact with their devices.

The dashboard summarizes key metrics such as:

- App usage time
- Screen-on time
- Number of installed apps
- Battery consumption
- Device model distribution
- User activity segmentation

# Key Metrics Summary

- Total Active Users: 700
- Top Device Model: Xiaomi Mi 11
- Average App Usage Time: 271 minutes per day
- Average Screen-On Time: 5 hours per day
- Average Number of Installed Apps: 51 apps
- Average Battery Drain: 1525 mAh per day

These metrics provide a baseline understanding of typical device engagement across the user base.

# Operating System Distribution

The dataset shows a clear dominance of Android users, representing approximately 79% of the user base, while iOS users account for 21%.

This distribution suggests that Android devices are significantly more common among the sampled users, which may influence overall engagement patterns and app usage behaviors.

Additionally, data usage across operating systems is relatively balanced:

- Android: ~51% of total data usage
- iOS: ~49% of total data usage

This indicates that although Android users are more numerous, iOS users demonstrate comparable data consumption levels, suggesting high engagement among iOS users.

# Device Model Insights

The analysis shows that Xiaomi Mi 11 and iPhone 12 are among the most active devices in terms of user count.

Other notable models include:

- Google Pixel 5
- Samsung Galaxy S21
- OnePlus 9

The Xiaomi Mi 11 leads in user representation, suggesting strong adoption or popularity within the dataset.

Understanding device distribution helps developers optimize applications for the most widely used devices.

# User Behavior Segmentation

Users were classified into three activity categories:

- Heavy Users – 40%
- Moderate Users – 39%
- Light Users – 21%

Heavy users represent the largest segment, indicating that a significant portion of users interact extensively with their devices throughout the day.

This segmentation is useful for:

- Identifying high-value users
- Targeting engagement campaigns
- Optimizing app performance for high-usage patterns

# App Installation Behavior

The analysis reveals clear differences in the average number of installed applications across user segments:

- Heavy Users: ~79 apps
- Moderate Users: ~50 apps
- Light Users: ~23 apps

Heavy users tend to explore and install more applications, which also correlates with higher screen time and app usage.

This suggests that app variety may drive increased engagement.

# Screen-On Time Patterns

Average screen-on time across devices ranges between 5.1 and 5.4 hours per day, indicating relatively consistent engagement across different device models.

Devices such as:

- iPhone 12
- Samsung Galaxy S21
- Xiaomi Mi 11

show slightly higher screen-on times, suggesting stronger daily usage.

# Battery Consumption Trends

Average daily battery consumption ranges from 1476 mAh to 1590 mAh depending on device model.

Devices with higher screen-on time tend to show higher battery drain, which suggests that device activity is the primary driver of energy usage.

This insight could help manufacturers optimize battery performance for high-engagement users.

# Demographic Behavior Patterns

User behavior also varies across age groups.

Younger users (18–34) tend to show a higher proportion of heavy usage patterns, while older age groups demonstrate more moderate or light usage.

This pattern aligns with typical digital engagement trends where younger users tend to spend more time interacting with mobile devices.

# Key Insights

Several important insights emerge from the analysis:

1. Android dominates the user base, but iOS users demonstrate comparable levels of data consumption.
2. Heavy users represent the largest segment, indicating strong overall engagement with mobile devices.
3. App installation count strongly correlates with usage intensity, with heavy users installing significantly more apps.
4. Battery consumption is largely driven by screen-on time and overall device activity.
5. Younger users are more likely to fall into the heavy usage category.

# Conclusion

This analysis highlights clear patterns in mobile device engagement across users, devices, and demographics.

Understanding these behavioral patterns can help:

- Mobile app developers optimize user experience
- Product teams prioritize features for high-engagement users
- Marketers design more targeted engagement strategies

Overall, the dashboard provides a comprehensive view of mobile device engagement trends and user behavior segmentation.